London's crime hotspots do not always "move".

But the answer changes when the measure changes.

In recent months, "crime hotspots" have dominated talk about London. Police have increased patrols in the West End and other hotspot areas. Around Christmas, Oxford Street businesses paid for extra security to deter phone snatching, pickpocketing, and shoplifting. A knife attack in Uxbridge in late October injured three people and killed one man; police said the suspect and victim did not know each other, and many read it as "random". When attention narrows to a few places, it is easy to assume the same hotspots keep returning.

At the same time, the UK's Office for National Statistics has said it will publish around 10% fewer outputs in 2026, and that some work on health and crime statistics may be scaled back as it prioritises "quality over quantity" and focuses on its core data.

Overseas voices have also dragged "crime hotspots" into the debate. Donald Trump and Elon Musk have described London as "out of control". The Financial Times pushed back and made a sharper point: people fear everyday crimes such as phone theft and knife-linked robbery and assault. Homicide rates can tell a different story, and London often looks less unusual in international comparisons. One city. Different measures. Different hotspot maps.

The story changes depending on what is counted. So before treating any hotspot map as "the truth", it helps to ask a basic follow-up: do the same places keep showing up as hotspots, or does the map shift over time and across crime types?

This brief asks one clear question: do London's crime hotspots stay in the same areas over time? It tests this in two ways. First, it compares the last 3 months with the last 12 months. Second, it keeps the same method but compares different crime types.

A simple rule for defining hotspots

This brief ranks London's 4,988 small areas (LSOAs) by crime rate per 1,000 residents. It then labels the top 20% as "hotspots" (Fig 1). This is a simple way to

capture a well-known pattern: crime tends to be concentrated in a relatively small number of places. (Sherman,1989).

For total crime over the last 12 months, the cut-off is 106.37 crimes per 1,000 residents a year. Areas above this line count as hotspots. The median area is about 63.39 per 1,000, so hotspots sit well above a typical neighbourhood (Fig 1).

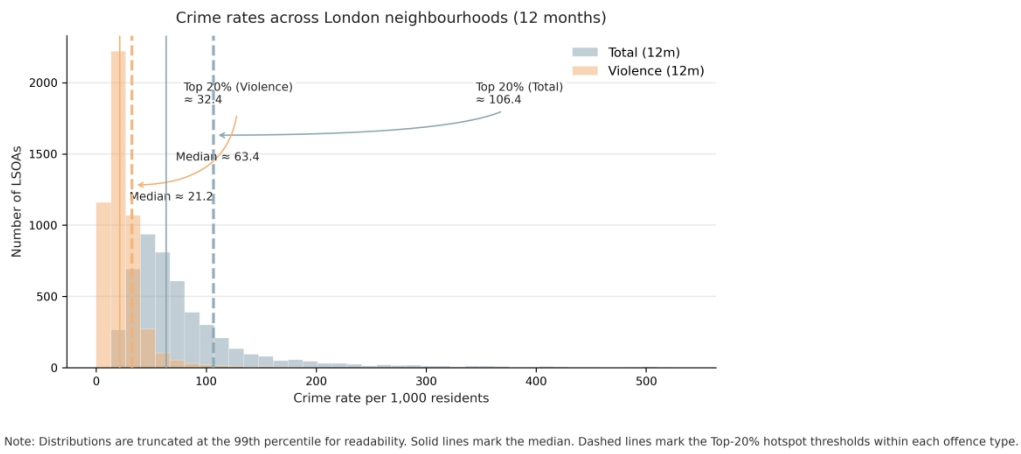

Fig 1. Crime rates across London's LSOAs

Distributions are truncated at the 99th percentile for readability. Solid lines mark the median. Dashed lines mark the Top-20% hotspot cut-offs within each offence type.

Fig 1 also shows that the same "top 20%" rule gives a different cut-off when the measure changes. Later, this brief tests whether the picture changes when the time window or crime type changes.

Angle 1: Total crime hotspots are mostly stable

Put the total-crime hotspot maps side by side (Fig 2a–b). The 12-month map and the 3-month map look very similar. Hotspots do not clearly "switch to a new set of places".

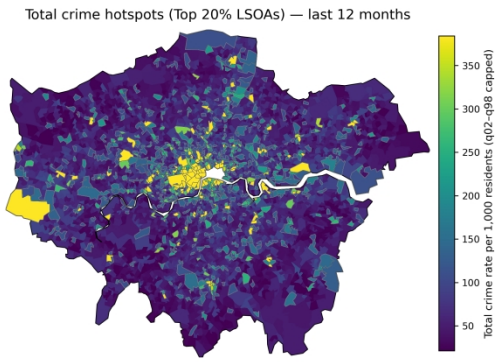

Fig 2a

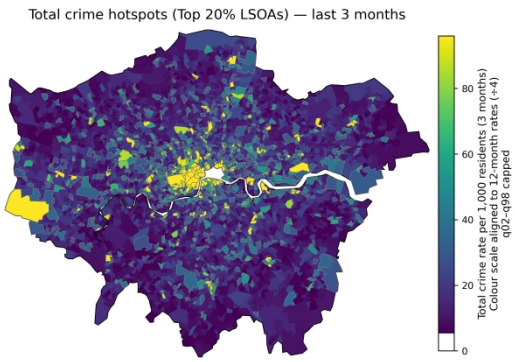

Fig 2b

Total crime hotspots (Top 20% LSOAs) for the last 12 months (top) and last 3 months (bottom), with the 3-month colour scale aligned to one quarter of the 12-month scale (÷4).

The numbers back this up. Of the 3-month hotspot areas, 87.4% are also hotspots in the 12-month view (overlap = 0.874). In plain terms, almost 9 in 10 places stay on the hotspot list.

The story is similar beyond the cut-off. This brief ranks all 4,988 areas by total crime; the 3-month and 12-month rankings line up closely. Think of two “top 100” charts from different periods: places near the top stay near the top (Spearman $\rho = 0.922$). Overall, the total-crime map looks stable, not like it reshuffles every few months.

Why the short window can “jump.”

A short window reacts faster. London’s 3-month rolling average moves up and down more than the 12-month average (Fig 3). The 12-month line shows the longer-run level. So small local changes in the 3-month hotspot map are not surprising. They may reflect short-term swings, not a real shift in the city’s overall pattern. This sets up the next step: Does the picture change when the crime type changes?

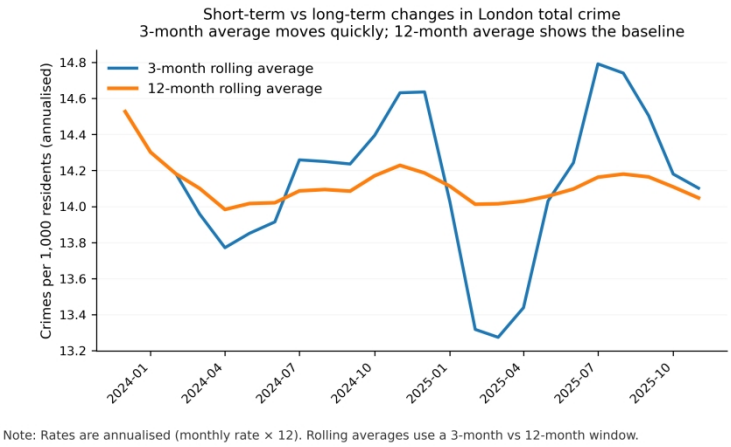

Fig 3. Short-term vs long-term change in London total crime

London's 3-month rolling average moves more than the 12-month average, showing how short windows pick up short-run swings rather than the longer-run level.

Why does this brief use violence against the person

This brief needs a comparison type that does not just mirror total crime. Theft is the largest category (33.8%), but it often tracks the total-crime pattern, so analysing theft alone adds little. Violence against the person is also a large share (26.7%), reflecting a different kind of risk, and sits at the centre of the safety debate, so the brief uses it as the comparison type.

Angle 2: Violence hotspots are less stable

Apply the same hotspot rule and time windows to violence. Compare the two maps (Fig 4a–b). Now the gap between the 12-month and 3-month maps is much bigger. Some places that are high over 12 months look different in the short window. The 3-month map also shows sharper local spikes.

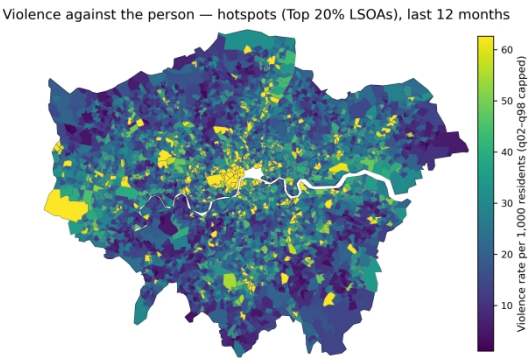

Fig 4a

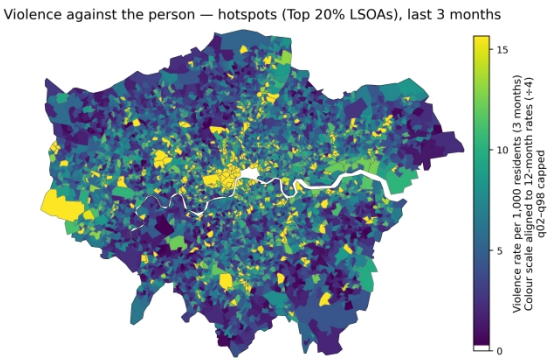

Fig 4b

Violence-against-the-person hotspots (top 20% LSOAs) over the last 12 months (top) and last 3 months (bottom); the 3-month map uses a colour scale aligned to the 12-month rates (÷4, q02–q98 capped) to aid visual comparison.

Think of it like a“top list”. Total crime looks more like a yearly list. Violence looks more like a short-term list: many places stay on it, but the order changes more often, and a few places jump up for a short time.

The numbers back this up. Among the 3-month violence hotspots, about 3 in 4 are also hotspots in the 12-month view (overlap = 0.747; Fig 5). The overall ordering is also fairly similar, but less so than for total crime (Spearman ρ = 0.855; Fig 5). Think of it like two class rankings: most places near the top stay near the top, but there is more swapping around.

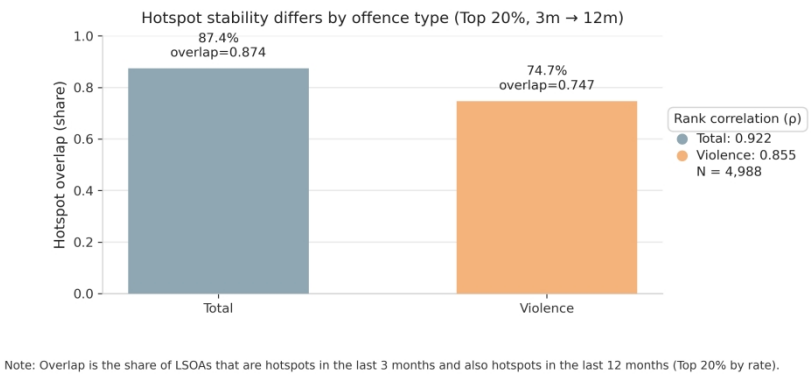

Fig 5. Hotspot stability differs by offence type

Hotspot persistence differs by offence type, with higher overlap between 3- and 12-month hotspots for total crime than for violence (top 20% LSOAs).

Conclusion

This brief asks one simple question: do London's crime hotspots stay in the same places over time? Mostly yes, but it depends on what is measured, so it helps to be clear about what "hotspot" means. For total crime, the 3-month and 12-month maps point to almost the same areas. For violence, the map does not flip, but it shifts more from one short period to the next. Claims that London is "out of control" often depend on what is counted, and different measures can point to different hotspot maps that steer patrols, security spending, and public attention. There are limits: the top-20% rule is a clear benchmark but still a choice, recorded crime can reflect reporting and policing, and this brief does not explain the sharp spikes in some places in the 3-month map, which may reflect rare incidents, reporting or policing effects, or a real local shift.

Appendix A: Methods and diagnostics

This appendix explains how I reached the main finding on whether crime hotspots stay in the same areas over time. It also includes key checks and spatial statistics for readers who want to verify the results.

1. Data and rate construction

I use two datasets: (1) Monthly recorded crime counts for London at LSOA level (including Major Category); (2) The ONS Mid-2024 population estimates (LSOA 2021). Because LSOAs vary greatly in population size, all comparisons use a crime rate per 1,000 residents.

I compute the rate as: rate = (crime count ÷ resident population) × 1,000. I use two time windows (the last 12 months and the last 3 months). For each window, I sum monthly crime counts within the window and then calculate the corresponding rate (units: crimes per 1,000 residents over that window).

I keep only areas that match across the crime data, population data, and the LSOA boundary file using LSOA Code (N = 4,988 LSOAs). A small number of areas appear blank on maps because some boundary geometries are missing or invalid. The main-text metrics (hotspot overlap and rank correlation) come from the matched table, so these mapping gaps do not change them.

In Angle 2, I use Violence Against the Person alongside total crime. In the study period, it accounts for 26.7% of recorded crimes (467,464 incidents), the second-largest major category after Theft (33.8%) (Fig A0). Unlike total crime, which combines many offences, a single offence type can vary more in the short run. This helps test whether hotspot stability depends on the metric used.

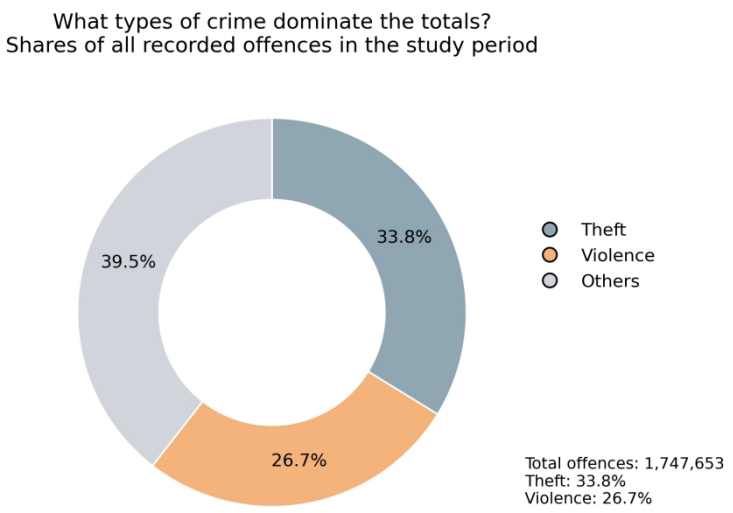

Fig A0. Offence composition

"Others" includes all major categories except Theft and Violence Against the Person.

2. Defining crime hotspots

In hotspot research, crime is not spread evenly across a city. A hotspot means that a small number of places account for a large share of crimes.[1]

To keep the definition clear and comparable across time windows, I rank all 4,988 London LSOAs by the chosen crime rate and label the top 20% as hotspots. This rank/quantile threshold approach is common in hotspot mapping and lets me compare results across windows and measures.[2]

For the last 12 months of total crime rate, the top-20% cut-off is 106.37 incidents per 1,000 residents per year, while the median LSOA is about 63.39 per 1,000. This shows that hotspot areas have much higher crime rates than a typical area.

Crime rates also show a clear long tail. Most areas sit in the middle of the distribution, but a small number of areas have very high values.[2]

For readability, some plots in the main text limit extreme values so that a few outliers do not flatten the rest of the chart. As a complement, Fig A1 uses a log-scaled frequency axis to show the same 12-month distribution, so the full tail remains visible.[2]

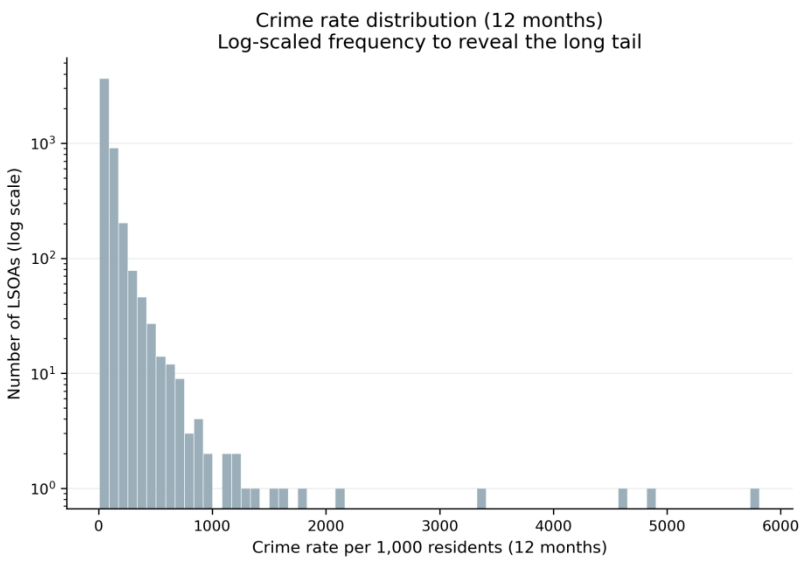

Fig A1. Distribution of total crime rates (12 months)

Distribution of total crime rate (last 12 months), with a log-scaled frequency axis to show the full tail.

3. Hotspot persistence metrics (Top-20% overlap + Spearman's ρ)

Using the matched set of N = 4,988 London LSOAs, I compute crime rates for the last 12 months and the last 3 months (crimes per 1,000 residents over each window). I summarise hotspot persistence with two measures: Top-20% overlap and Spearman's rank correlation (ρ).

Top-20% overlap treats the top 20% of LSOAs as hotspots and reports the share of 3-month hotspots that are also hotspots in the 12-month window. Spearman's ρ reports how similar the full LSOA rankings are between the two windows.

Results: Total crime ρ = 0.922, overlap = 0.874; Violence ρ = 0.855, overlap = 0.747.

To balance clarity and detail, Figs A2 and A3 show two scatterplots, each of which includes a full-range view and a zoomed (P99) view. The full-range plot keeps extreme values to show the long tail and overall range. The zoomed plot focuses on the main cloud, so typical patterns are easier to see. Both views use the same data and the same reference line; only the axis limits change.[3]

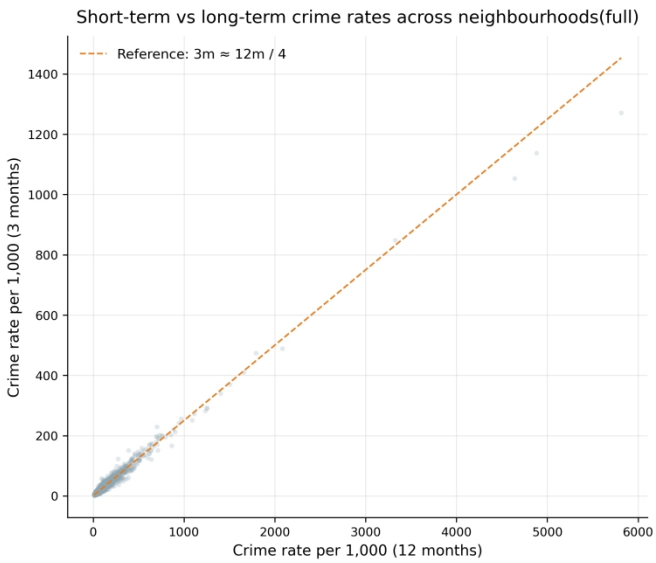

Fig A2a. Total crime (full range)

Scatter of LSOA crime rates per 1,000: 12-month (x-axis) vs 3-month (y-axis). The dashed line shows the reference 3m ≈ 12m/4.

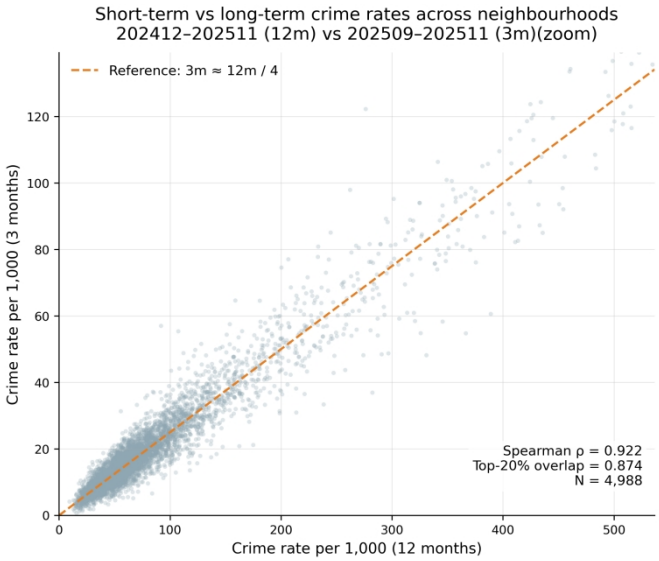

Fig A2b. Total crime (P99 zoom)

Same as Fig A2a, but axes are capped at the 99th percentile (P99) to make the main point cloud easier to see. The data and reference line are unchanged.

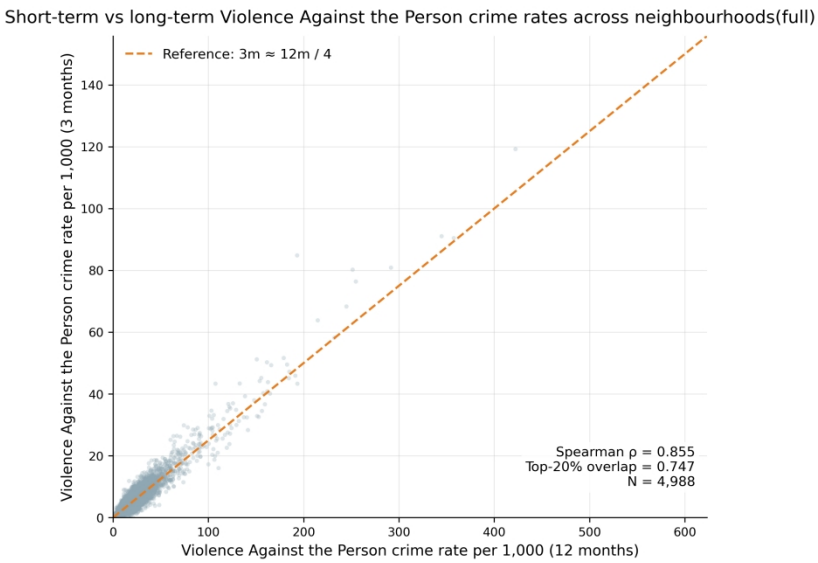

Fig A3a. Violence against the person (full range)

Scatter of LSOA crime rates per 1,000: 12-month (x-axis) vs 3-month (y-axis). The dashed line shows the reference 3m ≈ 12m/4.

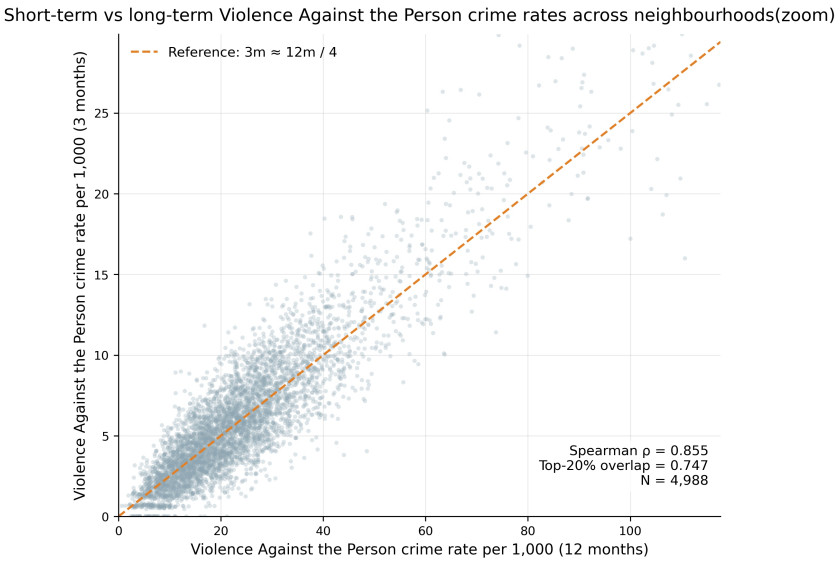

Fig A3b. Violence against the person (P99 zoom)

Same as Fig A3a, but axes are capped at P99 to improve readability. The data and reference line are unchanged.

4. Spatial autocorrelation: do hotspots "cluster" (Global Moran's I + LISA)

Neighbouring areas are not independent, so I test whether the 12-month total crime rate is spatially clustered.[4]

I build a Queen contiguity weights matrix (areas sharing an edge or a corner are neighbours) and apply row-standardisation.

Table A1 shows Global Moran's I = 0.514 with a permutation test p = 0.001 (N = 4,988).

This shows clear positive spatial autocorrelation: high-rate areas tend to sit near other high-rate areas, and low-rate areas near other low-rate areas. So hotspots tend to form clusters, not a few isolated points.

A global test does not show where clustering happens.[4]

I therefore use Local Moran's I (LISA, $p < 0.05$) to classify each LSOA as HH (high–high), LL (low–low), HL, LH, or not significant.[5]

Table A2 shows HH = 307 (6.2%) and LL = 651 (13.1%). HL = 59 (1.2%) and LH = 134 (2.7%) are rarer, and most areas are not significant (3,837; 76.9%). Here, HH indicates hotspot clusters, LL low-rate clusters, and HL/LH spatial outliers (an area differs from its neighbours).[5]

Fig A4 mirrors Table A2 on the map: HH forms contiguous patches, LL is more common in outer low-rate areas, and the other types are scattered, consistent with the Global Moran's I result.

In the previous section, I reported two persistence metrics (Top-20% overlap and Spearman's ρ) for total crime and violence. They suggest hotspots are more stable for total crime than for violence. This section adds one spatial point: hotspots often come in clusters, not single areas. Together, these checks support the brief's main message: hotspots mostly stay in the same places, but the picture depends on the measure and the time window.

Table A1. Global Moran's I (12-month total crime rate)

I = 0.514 | p = 0.0010 | N = 4,988

Item	Value
Variable	12-month total crime rate
Unit	Crimes per 1,000 residents
Spatial weights	Queen contiguity (share edge or corner)
Weight standardisation	Row-standardised (R)
Statistic	Global Moran's I
Result	**I = 0.514**
Permutation p-value	**p = 0.0010**
Sample size (N)	4,988

Table A1. Global spatial autocorrelation (12-month total crime rate)

Table A2. Local clusters (LISA, p < 0.05) — 12-month total crime

N = 4,988

Cluster	Count	Share
High-High	307	6.2%
Low-Low	651	13.1%
High-Low	59	1.2%
Low-High	134	2.7%
Not significant	3837	76.9%

Table A2. Local cluster types (LISA, p < 0.05; 12-month total crime).

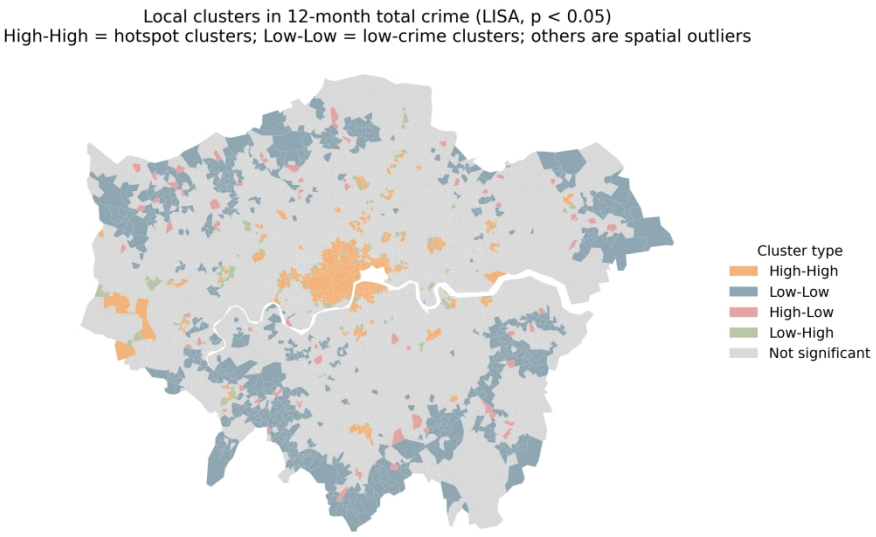

Fig A4. LISA cluster map (12-month total crime)

This figure provides a visual check against Table A2 and shows whether hotspots appear as contiguous clusters.

5. Limitations and interpretation

This analysis uses one main crime measure and one hotspot rule, so the findings are measure-specific rather than a claim about all offences. The top-20% cut-off is a transparent benchmark used in the literature, but it is still a choice: small changes can move borderline areas across the line. Recorded crime also reflects reporting and policing, and short windows can shift because of seasonality or noise, so apparent "movement" should be interpreted cautiously. Finally, some areas are missing from the maps due to join/geometry issues that I do not fully diagnose here, so the visuals should be read as indicative patterns rather than exact boundaries.

References

[1] Sherman, L.W. (1995) 'Hot Spots of Crime and Criminal Careers of Places', in Eck, J.E. and Weisburd, D. (eds.) *Crime and Place* (Crime Prevention Studies, Vol. 4), pp. 35–52. Monsey, NY: Criminal Justice Press. https://popcenter.asu.edu/content/crime-prevention-studies-volume-4-volume-4

[2] Cameron, J.G. and Leitner, M. (2005) 'Spatial Analysis Tools for Identifying Hot Spots', in Eck, J.E., Chainey, S., Cameron, J.G., Leitner, M. and Wilson, R.E. *Mapping Crime: Understanding Hot Spots* (NIJ Special Report, NCJ 209393). Washington, DC: National Institute of Justice. (Chapter 3). https://nij.ojp.gov/library/publications/mapping-crime-understanding-hot-spots

[3] Wilke, C.O. (2019) *Fundamentals of Data Visualization*. https://clauswilke.com/dataviz/ecdf-qq.html

[4] Rey, S.J., Arribas-Bel, D. and Wolf, L.J. (2023) 'Global Spatial Autocorrelation', in *Geographic Data Science with Python*. https://geographicdata.science/book/notebooks/06_spatial_autocorrelation.html

[5] Rey, S.J., Arribas-Bel, D. and Wolf, L.J. (2023) 'Local Spatial Autocorrelation', in *Geographic Data Science with Python*. https://geographicdata.science/book/notebooks/07_local_autocorrelation.html